Implementation Roadmap

EMS Readiness Optimization — Manhattan Deployment Plan

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Overview

This roadmap describes the phased deployment of the demand-weighted optimized ambulance allocation policy (P2) across Manhattan's 48 FDNY firehouses. The plan is designed to be low-risk, evidence-based, and incrementally validated.

Phase 1: Pilot Deployment (Months 1–3)

Objective

Validate P2 allocation at 5 high-impact firehouses and confirm simulation predictions with real-world data.

Scope

- **Firehouses**: 5 highest-impact locations (selected based on P2 allocation recommendations for precincts with highest demand)
- **Units**: 5–8 ambulances repositioned according to P2 staging plan
- **Duration**: 90 days

Activities

Week	Activity	Owner
1–2	Briefing for pilot firehouse captains and dispatch supervisors	Operations
1–2	Install KPI monitoring dashboard (mean RT, 8-min coverage, utilization)	IT/Analytics
3	Begin P2 staging at pilot firehouses	Operations
3–12	Continuous monitoring and weekly performance reviews	Analytics
8	Mid-pilot assessment and adjustment	Leadership
12	Pilot completion report with go/no-go for Phase 2	Research Team

Success Criteria

- ≥15% improvement in mean response time in pilot precincts
- ≥10 percentage point improvement in 8-min coverage in pilot area
- No degradation in non-pilot area response times >5%
- Positive feedback from dispatch supervisors and crews

Resources Required

- 0 additional ambulances (repositioning only)
- 1 analytics dashboard (estimated 2 weeks development)
- Training: 2-hour briefing per pilot firehouse
- Monitoring: 4 hours/week analytics staff time

Risk Mitigation

Risk	Probability	Impact	Mitigation
Crew resistance to new staging	Medium	Medium	Early engagement, explain rationale with data
Unexpected demand pattern shift	Low	Medium	Weekly monitoring, pre-defined rollback criteria
Adjacent area performance degradation	Low	High	Monitor non-pilot areas; rollback if >5% degradation

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Phase 2: Gradual Rollout (Months 3–6)

Objective

Expand P2 allocation to 15–20 firehouses based on pilot learnings, and calibrate the model with real dispatch data.

Scope

- **Firehouses**: 15–20 (pilot 5 + 10–15 additional)
- **Units**: 12–16 ambulances under P2 staging
- **Duration**: 90 days

Activities

Month	Activity	Owner
3	Analyze pilot results; calibrate travel time model with real data	Research
3	Select Phase 2 firehouses based on expected impact	Research + Ops
4	Training sessions for Phase 2 firehouse crews	Operations
4–6	Staged rollout (5 new firehouses per 2 weeks)	Operations
4–6	Continuous monitoring with expanded dashboard	Analytics
5	Integrate road-network travel times (OSRM)	IT/Research
6	Phase 2 completion report	Research Team

Model Calibration

- Compare predicted vs actual response times from pilot data
- Adjust speed parameters based on real dispatch records
- Re-optimize P2 allocation if significant calibration changes

Success Criteria

- Consistent ≥30% improvement in mean RT across expanded area
- ≥90% 8-min coverage across P2-served precincts
- Model predictions within ±15% of observed response times
- No reported operational issues from dispatch

Phase 3: Full Deployment (Months 6–12)

Objective

Deploy P2 allocation across all 48 Manhattan firehouses with shift-specific optimization.

Scope

- **Firehouses**: All 48 Manhattan firehouses
- **Units**: Full fleet under P2 staging
- **Duration**: 6 months (ongoing operations thereafter)

Activities

Month	Activity	Owner
6–7	Complete rollout to remaining firehouses	Operations
7–8	Develop shift-specific allocations (day/evening/overnight)	Research
8–9	Implement time-of-day P2 variants	Operations
9–10	Final model validation with 6 months of operational data	Research
10–12	Documentation and handoff to operations team	Research
12	Final deployment report and future recommendations	Research

Advanced Features

1. **Shift-specific staging**: Different P2 allocations for day (8AM–4PM), evening (4PM–12AM), and overnight (12AM–8AM) shifts
2. **Weekly adjustment**: Incorporate day-of-week demand patterns
3. **Seasonal calibration**: Quarterly model re-calibration

Timeline Summary

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Month:  1    2    3    4    5    6    7    8    9    10   11   12
        |----Phase 1 (Pilot)----|
                                |------Phase 2 (Expansion)------|
                                                  |------Phase 3 (Full)------|
        [Monitoring Dashboard Development]
                          [Road Network Integration]
                                              [Shift-Specific Optimization]
                                                                [Handoff & Documenta-
tion]
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Resource Requirements

Personnel

Role	FTE	Duration	Notes
Data Analyst	0.5	12 months	Dashboard, monitoring, calibration
Operations Coordinator	0.25	12 months	Firehouse liaison, training
Research Lead	0.25	12 months	Model calibration, optimization updates
IT Support	0.1	3 months	Dashboard deployment

Technology

Component	Cost Estimate	Timeline
KPI Dashboard (web-based)	$15–25K development	Month 1–2
OSRM routing server (optional)	$5K/year hosting	Month 4–5
Data pipeline (CAD integration)	$10–20K development	Month 6–8

Training

- Phase 1: 5 briefings × 2 hours = 10 hours
- Phase 2: 15 briefings × 1.5 hours = 22.5 hours
- Phase 3: 28 briefings × 1 hour = 28 hours
- Materials: Staging maps, quick-reference cards, FAQ documents

Success Metrics and Monitoring

Key Performance Indicators

KPI	Baseline (P0)	Target	Measurement
Mean Response Time	8.08 min	< 3.5 min	Daily from CAD
8-min Coverage	64.4%	> 95%	Weekly calculation
P90 Response Time	19.47 min	< 5 min	Weekly calculation
Utilization	9.1%	7–10%	Weekly calculation
Crew satisfaction	N/A	> 80% positive	Quarterly survey

Monitoring Cadence

- **Daily**: Mean RT and 8-min coverage
- **Weekly**: Full KPI dashboard review; anomaly detection
- **Monthly**: Trend analysis and model performance assessment
- **Quarterly**: Comprehensive review with leadership; model recalibration

Rollback Criteria

- If mean RT increases by >20% in any precinct for >2 consecutive weeks
- If 8-min coverage drops below 50% in any precinct
- If dispatch supervisors report critical operational issues
- Rollback procedure: Revert to P0 staging at affected firehouses within 24 hours

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Technology Integration Needs

Phase 1 (Dashboard Only)

- Web-based dashboard displaying real-time KPIs
- Data source: Manual extraction from CAD system
- Hosting: Internal server or cloud (Heroku/AWS)

Phase 2 (Routing Integration)

- OSRM or similar open-source routing engine
- Real road-network travel time calculations
- Replaces Haversine-based proxy for improved accuracy

Phase 3 (CAD Integration)

- Automated data feed from Computer-Aided Dispatch
- Real-time unit location tracking
- Automated P2 allocation recommendations at shift change
- Future: Dynamic repositioning suggestions

Risk Register

ID	Risk	Probability	Impact	Mitigation	Owner
R1	Crew resistance	Medium	Medium	Early communication, data-driven rationale	Operations
R2	Model prediction error	Low	Medium	Calibration with real data; gradual rollout	Research
R3	Adjacent area degradation	Low	High	Monitor all precincts; rollback trigger	Analytics
R4	Demand pattern change	Low	Low	Quarterly recalibration; adaptive model	Research
R5	Technology delays	Medium	Low	Phase 1 requires no technology; decouple	IT
R6	Leadership change	Low	Medium	Document rationale; institutional knowledge	Research

Budget Summary

Category	Phase 1	Phase 2	Phase 3	Total
Personnel	$25K	$25K	$50K	$100K
Technology	$20K	$10K	$15K	$45K
Training	$2K	$3K	$5K	$10K
Contingency (15%)	$7K	$6K	$10.5K	$23.5K
Total	**$54K**	**$44K**	**$80.5K**	**$178.5K**

Note: Costs are estimates. No new ambulances are required — the investment is in analytics, training, and monitoring.

Expected Return on Investment

- **Annual time saved**: ~167,750 minutes (2,796 ambulance-hours)
- **Equivalent fleet expansion**: 3× effective capacity
- **If fleet could be reduced by even 2 units**: $200K+/year savings in operating costs
- **Payback period**: < 12 months based on conservative efficiency gains

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Prepared by the EMS Optimization Research Team. For questions, contact the project lead or refer to the full technical report (docs/technical_report.md).